Grade 12 Practice

Standard S.ID.3

Computation

1. Adding an outlier affects which measure the most? Write mean or median. **mean**

2. Does the median resist outliers? Write yes or no. **yes**

3. Data are 1, 2, 3, 4. If 100 is added, does the mean increase? Write yes or no. **yes**

4. Does an outlier affect the range? Write yes or no. **yes**

Word problems

5. Salaries include one very high CEO salary. Which center should be reported to show a typical salary? Write mean or median.
Answer: median

6. Data are 2, 3, 4, 5, and 50 is added. What is the new range?
Answer: 48

Grade 12 Practice

Standard S.ID.4

Computation

1. A normal distribution has mean 50 and standard deviation 10. What is the z-score of

2. What is the z-score of 40 in the same distribution? ___ **-1**

3. In a normal distribution, about what percent of data falls within one standard deviat

4. What is the z-score of the mean? ___ **0**

Word problems

5. A test has mean 70 and standard deviation 5. What is the z-score of a score of 80?
Answer: 2

6. Heights are normal with mean 65 inches and standard deviation 3 inches. About what percent lie within one standard deviation?
Answer: 68%

Grade 12 Practice

Standard S.IC.1

Computation

1. A census surveys every member of a population. Write yes or no: is this a sample? **no**

2. A survey asks 100 students from a school of 1,000. Is this a sample? Write yes or no. **y**

3. Is a sample a subset of a population? Write yes or no. **yes**

4. A parameter describes a population. What does a statistic describe? Write your answ

Word problems

5. A poll of 500 voters is taken in a city of 80,000. What is the sample size?
Answer: 500 voters

6. The average lifetime of all bulbs a factory makes is a ___. Write parameter or statistic.
Answer: parameter

Grade 12 Practice

Standard S.IC.2

Computation

1. A fair coin is modeled with 2 equal outcomes. Is this a valid model for a 50% chance?

2. A 10% chance is modeled with 10 equal outcomes, where 1 counts as success. Is this v

3. A simulation results are compared to the theoretical probability. Is this a valid check

4. A 1/4 chance is modeled with 4 equal outcomes, where 1 counts as success. Is this v

Word problems

5. A spinner has 3 equal sections. Is it a valid model for a 1/3 chance? Write yes or no.
Answer: yes

6. A player makes 70% of shots. Is a model with 10 equal outcomes, where 7 count as success, valid? Write yes or no.
Answer: yes

Grade 12 Practice

Standard S.IC.3

Computation

1. Is a random sample unbiased? Write yes or no. **yes**

2. Surveying only students who ride the bus is biased. Write yes or no. **yes**

3. Is a convenience sample random? Write yes or no. **no**

4. Does a stratified sample use groups? Write yes or no. **yes**

Word problems

5. A teacher picks every 10th name from an alphabetical list. What sampling method is this?
Answer: systematic

6. A survey picks 20 students from each grade level. What sampling method is this?
Answer: stratified

Grade 12 Practice

Standard S.IC.4

Computation

1. A sample of size 100 has a margin of error of about how many percent? ___ **10%**

2. Does the margin of error decrease when sample size increases? Write yes or no. **yes**

3. A sample of size 400 has a margin of error of about how many percent? ___ **5%**

4. A confidence interval is written as estimate ± ___. **margin of error**

Word problems

5. A poll shows 60% support with a margin of error of 3%. What is the confidence interval?
Answer: 57% to 63%

6. A sample of 1,000 has a margin of error of about how many percent?
Answer: about 3%

Grade 12 Practice

Standard S.IC.5

Computation

1. Does random assignment help show cause? Write yes or no. **yes**

2. What is the purpose of a control group? Write your answer in one word. **comparison**

3. If a difference is larger than chance would explain, is it significant? Write yes or no. **y**

4. What is the purpose of a placebo? Write blinding or placebo effect control. **blinding**

Word problems

5. A treatment group improves by 8 points and a control group improves by 2 points. What is the difference?
Answer: 6 points

6. Without random assignment, can you prove causation? Write yes or no.
Answer: no

Grade 12 Practice

Standard S.IC.6

Computation

1. If a sample is biased, are the conclusions limited? Write yes or no. **yes**

2. Does correlation prove causation? Write yes or no. **no**

3. Is a sample of 10 people always representative of a city? Write yes or no. **no**

4. Is a larger random sample generally more reliable? Write yes or no. **yes**

Word problems

5. A report says a drug works based on 20 people with no control group. Is it strong evidence? Write yes or no.
Answer: no

6. A survey was only done online. Can it be used to describe all adults? Write yes or no.
Answer: no

Grade 12 Practice

Standard S.CP.1

Computation

1. How many outcomes are in the sample space for rolling one die? ___ **6**

2. How many outcomes are in the event 'even' when rolling a die? ___ **3**

3. If $P(A) = 0.3$, what is the probability of the complement of A? ___ **0.7**

4. How many outcomes are in the sample space for flipping two coins? ___ **4**

Word problems

5. How many cards in a standard deck are red? (Event: red card.)
Answer: 26

6. When one die is rolled, the event 'roll greater than 4' has how many outcomes?
Answer: 2

Grade 12 Practice

Standard S.CP.2

Computation

1. $P(A) = 0.5$, $P(B) = 0.4$, and $P(A \text{ and } B) = 0.2$. Are A and B independent? Write yes or no.

2. $P(A) = 0.3$, $P(B) = 0.6$, and $P(A \text{ and } B) = 0.1$. Are A and B independent? Write yes or no.

3. Are two coin flips independent? Write yes or no. **yes**

4. Are two draws without replacement independent? Write yes or no. **no**

Word problems

5. Whether it rains and whether a sunny day occurs are independent? Write yes or no.
Answer: no

6. Two dice are rolled. Are the results independent? Write yes or no.
Answer: yes

Grade 12 Practice

Standard S.CP.3

Computation

1. P(A and B) = 0.2 and P(B) = 0.5. What is P(A | B)? ___ **0.4**

2. P(A and B) = 0.09 and P(A | B) = 0.3. What is P(B)? ___ **0.3**

3. If A and B are independent, P(A | B) equals ___. **P(A)**

4. Write the formula for P(A | B). **P(A and B) / P(B)**

Word problems

5. Of 15 students who play sports, 10 are girls. What is P(girl | plays sports)? Write as a fraction.
Answer: 2/3

6. P(B) = 0.25 and P(A and B) = 0.05. What is P(A | B)?
Answer: 0.2

Grade 12 Practice

Standard S.CP.4

Computation

1. A survey of 100 people has Group A with 30 yes and 20 no, and Group B with 10 yes ar

2. What is P(yes | Group A)? Write as a fraction. **3/5**

3. What is P(yes | Group B)? Write as a fraction. **1/5**

4. What is P(Group A)? Write as a fraction. **1/2**

Word problems

5. Which group is more likely to say yes? Write A or B.
Answer: A

6. What is P(Group A and no)? Write as a fraction.
Answer: 1/5

Grade 12 Practice

Standard S.CP.5

Computation

1. P(A | B) means the probability of A given that ___ happened. **B**

2. Two independent events both occur. Their probabilities are multiplied. Write yes or n

3. P(A) = 0.5 and P(B | A) = 0.2. What is P(A and B)? ___ **0.1**

4. P(B | A) = 0.2 and P(B) = 0.2. Are A and B independent? Write yes or no. **yes**

Word problems

5. Events A and B are independent. P(A) = 0.3 and P(B) = 0.4. What is P(A and B)?
Answer: 0.12

6. Two components each work with probability 0.9. What is the probability both work?
Answer: 0.81

Grade 12 Practice

Standard S.CP.6

Computation

1. Of 200 cars, 50 have GPS and 40 of those have a radio. What is P(radio | GPS)? **0.8**

2. What is P(GPS) out of 200 cars? Write as a fraction. **1/4**

3. What is P(GPS and radio) out of 200 cars? Write as a fraction. **1/5**

4. Radio is in 80 of the 200 cars. Are GPS and radio independent? Write yes or no. **no**

Word problems

5. A bag has 5 red and 3 blue marbles. Two are drawn without replacement. What is P(red then blue)? Write as a fraction.
Answer: 15/56

6. The same bag is used, but the first marble is replaced. What is P(red then blue)? Write as a fraction.
Answer: 15/64

Grade 12 Practice

Standard S.CP.7

Computation

1. $P(A) = 0.4$, $P(B \mid A) = 0.5$, and $P(B) = 0.25$. What is $P(A \mid B)$? **0.8**

2. $P(A) = 0.2$ and $P(B \mid A) = 0.5$. What is $P(A \text{ and } B)$? **0.1**

3. If $P(B \mid A) = P(B)$, are the events independent? Write yes or no. **yes**

4. $P(A) = 0.6$ and $P(B \mid A) = 0.1$. What is $P(A \text{ and } B)$? **0.06**

Word problems

5. $P(A) = 0.3$, $P(B \mid A) = 0.4$, and $P(B) = 0.2$. What is $P(A \mid B)$?
Answer: 0.6

6. A spam filter flags 90% of spam. 20% of email is spam, and 25% of email is flagged. What is $P(\text{spam} \mid \text{flagged})$?
Answer: 0.72

Grade 12 Practice

Standard S.MD.1

Computation

1. A game pays $10 with probability 0.1 and $0 otherwise. What is the expected value? $

2. A game pays $5 with probability 1/2 and costs $1 otherwise. What is the expected val

3. What is the expected value of a fair game? ___ **0**

4. What is the expected value of rolling a fair die and winning the number shown? ___

Word problems

5. A lottery ticket costs $2 and pays $100 with probability 1/100. What is the expected net value?
Answer: -$1

6. A random number from 1 to 4 pays that many dollars. What is the expected payout?
Answer: $2.50

Grade 12 Practice

Standard S.MD.2

Computation

1. A random variable X is 1, 2, or 3, each with probability 1/3. What is E(X)? **2**

2. X is 0 with probability 0.5 and 4 with probability 0.5. What is E(X)? **2**

3. X is 10 with probability 0.2 and 0 with probability 0.8. What is E(X)? **2**

4. X is 3 with probability 0.25 and 1 with probability 0.75. What is E(X)? **1.5**

Word problems

5. A store sells 0, 1, or 2 items with probabilities 0.5, 0.3, and 0.2. What is the expected number of items?
Answer: 0.7 items

6. A game pays $20 with probability 0.1 and costs $2 otherwise. What is the expected value?
Answer: $0.20

Grade 12 Practice

Standard S.MD.3

Computation

1. Do the probabilities of a distribution have to sum to 1? Write yes or no. **yes**

2. Are probabilities 0.2, 0.3, and 0.5 a valid distribution? Write yes or no. **yes**

3. Are probabilities 0.4, 0.4, and 0.4 a valid distribution? Write yes or no. **no**

4. If $P(0) = 0.2$ and $P(1) = 0.3$, what is $P(X \leq 1)$? ___ **0.5**

Word problems

5. A fair die is modeled with $P(X = 1)$ through $P(X = 6)$. What do the six probabilities sum to?
Answer: 1

6. Probabilities for 0, 1, and 2 are 0.5, 0.3, and x. What is x?
Answer: 0.2

Grade 12 Practice

Standard S.MD.4

Computation

1. When choosing between two options, should you pick the one with higher expected v

2. Option A has an expected value of 10 and option B has 8. Which should you choose?

3. A decision is based only on expected value. Is this always the best choice? Write yes

4. Expected value 0 means the game is what? Write fair or unfair. **fair**

Word problems

5. Option A has an expected profit of $500. Option B has $450. Which option should a company pick?
Answer: A

6. Two games have expected values of -$0.50 and $0. Which game should you choose?
Answer: the $0 game

Grade 12 Practice

Standard S.MD.6

Computation

1. Is a game with expected value 0 fair? Write yes or no. **yes**

2. A game wins $4 with probability 0.2 and loses $1 with probability 0.8. Is it fair? Write y

3. A spinner has 4 equal sections, each with a prize. Is the spinner fair? Write yes or no.

4. A game has expected value -0.25. Is it in your favor? Write yes or no. **no**

Word problems

5. A dice game pays $6 on a 6 and costs $1 otherwise, with probabilities 1/6 and 5/6. Is it fair? Write yes or no.
Answer: no

6. You pay $5 to enter and win $10 with probability 0.5. Is it fair? Write yes or no.
Answer: yes